

JASWANTH KUMAR

Data Scientist | Machine Learning Engineer| +1 551-260-5670 | jaswanthk1006@gmail.com

SUMMARY

- Results-driven Data Scientist with 4+ years of experience architecting and deploying AI-powered products that drive strategic business growth. Expert in translating complex business challenges into scalable technical solutions, with a proven track record in leading the development of Generative AI and RAG platforms that empower cross-functional teams. Combines deep technical expertise in LLMs, NLP, and MLOps with a strong product mindset to deliver measurable improvements in efficiency, decision-making, and client engagement.

EXPERIENCE

Unum, USA | Data Scientist Feb 2024 - Current

- Applied expertise in data structures and algorithms to architect high-performance, scalable data pipelines and ML inference services, optimizing for efficiency and low-latency (<100ms) response times in Python.
- Designed Spearheaded the vision and strategy for core **products** (LLM fine-tuning service, RAG-as-a-Service), defining the **long-term roadmap** and prioritizing development to meet internal user needs, which **boosted team experimentation by 30%**.
- Partnered with **stakeholders across the organization** to identify opportunities, leveraging customer data and **predictive modeling to drive business solutions** and **generate revenue** via enhanced marketing targeting and insight tools.
- Collaborated with data scientists to **set technical strategy**, architecting and implementing solutions for **recommendation systems** and **time series forecasting** challenges using Python.
- Engineered and deployed scalable, high-availability services on AWS that reliably served **50,000+ daily predictions** with sub-100ms latency, supporting critical business operations.
- Established robust **A/B testing** frameworks to rigorously validate model performance and quantify business impact, ensuring data-driven decision-making.
- Researched and integrated **emerging technologies** like vector databases (Pinecone) and LLM orchestration (LangChain) to maintain a technological competitive edge and accelerate development.
- Partnered cross-functionally with product and engineering teams to align ML-driven solutions with business objectives.

Infinite Infolab, India | Machine Learning Engineer Mar 2021 - Jul 2023

- Leveraged **Power BI** to develop interactive dashboards that visualized real-time model results, allowing stakeholders to easily validate insights and ensure alignment with business objectives prior to full-scale deployment.
- Collaborated with **data engineers** to design and optimize **ETL processes**, leveraging **SQL** to extract, clean, and transform data for analysis, ensuring efficient data flow, improved data quality, and streamlined processes for data-driven decision-making.
- Designed and implemented **end-to-end NLP pipelines** (text classification, clustering, semantic similarity, entity extraction) using **spaCy, NLTK, and Hugging Face Transformers**.
- Utilized **NLP techniques** and **NLTK (Natural Language Toolkit)** for text preprocessing, feature extraction, and text mining tasks, transforming unstructured data into structured datasets, enabling more effective analysis and actionable insights across projects.
- Built and validated machine learning models, including **Logistic Regression, Decision Trees, and Neural Networks**, to predict key business outcomes such as **customer churn** and **product recommendations**, driving strategic decisions and business growth.
- Applied **time series analysis** techniques to analyze sequential data, identify trends, and implement both supervised and unsupervised learning algorithms, improving forecasting accuracy and uncovering hidden patterns in the data.
- Managed and optimized **data workflows** in **Amazon Redshift**, utilizing its **data warehousing capabilities** to efficiently process and analyze large datasets, improving **query performance** and significantly reducing **data processing times** across operations.
- Deployed and managed **cloud-based infrastructure** on AWS, including **EC2 instances, S3 buckets, RDS databases, and Lambda functions**, ensuring system **scalability, high availability**, and enhanced **performance** across the environment.
- Actively participated in agile team retrospectives and knowledge-sharing sessions** to improve project workflows and mentor junior colleagues.

TECHNICAL SKILLS

Language: Python, R, SQL, Shell Scripting
IDEs: Visual Studio Code, PyCharm, Jupyter Notebook, Google Colab
Statistical Methods: Hypothesis Testing, ANOVA, Time Series, A/B Testing
Machine Learning: Regression analysis, Bayesian Method, Decision Tree, Random Forests, Neural Network, Gen AI, RAG Sentiment Analysis, K- Means Clustering, KNN, Classification, SVM, Natural Language Processing (NLP), LLM, CNN, OpenCV.
Packages: NumPy, Pandas, Matplotlib, SciPy, Scikit-Learn, PyTorch, TensorFlow, Keras, LangChain, Spark, Seaborn,
Visualization Tools / Database: Tableau, Power BI, Microsoft Excel, MySQL, SQL Server, Oracle, MongoDB, Vector DB
Cloud Technologies: AWS, Azure
Software/Other Skills: Jira, Data Cleaning, Data Wrangling, Critical Thinking, Communication Skills, Presentation Skills, Problem-solving, Decision-Making, EDA, Communication Skills, Databricks, Data Visualization, Predictive Analytics, Pattern Recognition, Data Integrity, Quantitative Data, Data Science, Statistics, Statistical Analysis, Data Analytics, Data Modeling, Big Query, Snowflake, SDLC, Agile, Waterfall, Figma
Operating System: Windows, Linux

EDUCATION

New Jersey Institute of Technology, Newark, NJ, USA.
Master's of Science in Data Science
Sagi Rama Krishnam Raju Engineering College India
Bachelor's of Technology in Computer Science and Engineering